

Welcome to Prism

What is Prism?

Prism helps developers understand the environmental impact of their AI usage.

As AI becomes a larger part of software development, it can be difficult to see how often AI is being used, which models are being selected, and what impact those choices have. Prism makes those interactions visible by estimating the carbon emissions associated with AI-powered development activities.

Prism is not designed to discourage the use of AI. Instead, it helps teams build awareness of their usage patterns and make more informed decisions about when and how AI is used.

What Prism Measures

Prism tracks AI usage in two areas.

Development Time

Interactions with AI tools that support software development.

Currently supported:

- GitHub Copilot
- Claude Code

Runtime

AI requests generated by applications while they are being developed, tested, and executed.

What Is Included

Prism focuses on operational emissions associated with AI inference during development.

What Is Not Included

Prism does not currently account for:

- Model training
- Hardware manufacturing
- Data centre construction
- Water consumption
- Wider system-level impacts



How Prism Works

Prism analyses AI token usage and applies model-specific conversion factors based on published research.

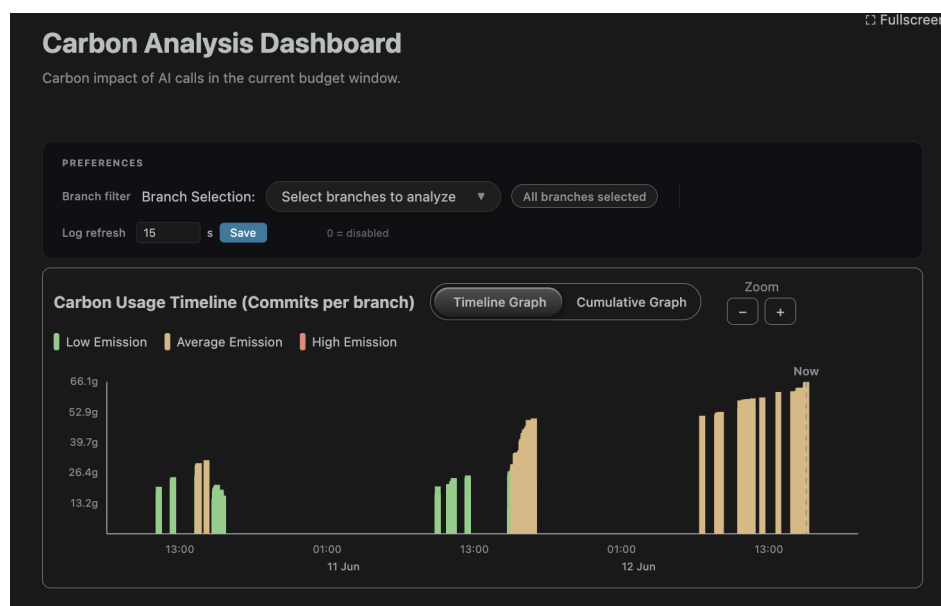
The resulting estimates are displayed through dashboards, logs, and visual indicators that help users understand how AI is being used within a project.

Because AI providers do not currently publish complete energy or emissions data, all values should be considered estimates rather than exact measurements.

For a detailed explanation of the methodology, assumptions, and data sources, refer to the Emission Calculation Methodology document.

Understanding the Dashboard

The dashboard provides a project-level view of AI activity and estimated impact across both development-time and runtime interactions.



Activity Log

The activity log records every captured interaction.

Each entry contains:

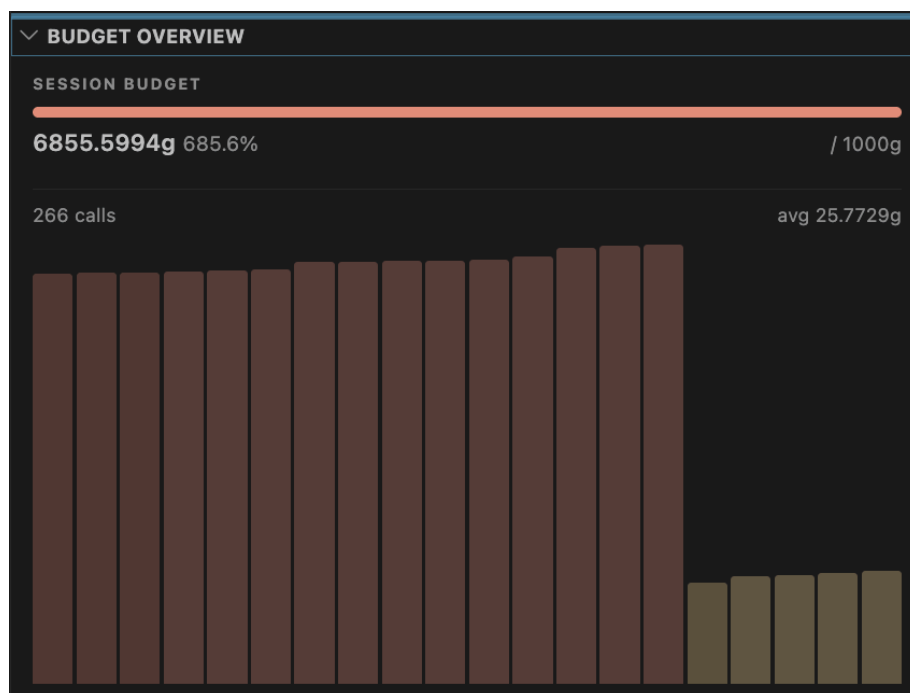
- Estimated emissions
- Model used
- Timestamp

Logs are displayed with the most recent requests at the top, making it easier to review current activity.

ESTIMATING CARBON: CARBON ESTIMATES			
Last 261 calls:			
66.1021g CO ₂ e	[Log · Claude Code]	claude-sonnet-4-6	12/06/2026, 16:02:58
66.0085g CO ₂ e	[Log · Claude Code]	claude-sonnet-4-6	12/06/2026, 16:02:07
65.5341g CO ₂ e	[Log · Claude Code]	claude-sonnet-4-6	12/06/2026, 16:01:03
64.2603g CO ₂ e	[Log · Claude Code]	claude-sonnet-4-6	12/06/2026, 16:00:37
63.8474g CO ₂ e	[Log · Claude Code]	claude-sonnet-4-6	12/06/2026, 16:00:31
63.7383g CO ₂ e	[Log · Claude Code]	claude-sonnet-4-6	12/06/2026, 15:58:12
63.6467g CO ₂ e	[Log · Claude Code]	claude-sonnet-4-6	12/06/2026, 15:58:08
63.5778g CO ₂ e	[Log · Claude Code]	claude-sonnet-4-6	12/06/2026, 15:30:37
63.5245g CO ₂ e	[Log · Claude Code]	claude-sonnet-4-6	12/06/2026, 15:24:30
62.2974g CO ₂ e	[Log · Claude Code]	claude-sonnet-4-6	12/06/2026, 15:24:22
62.1419g CO ₂ e	[Log · Claude Code]	claude-sonnet-4-6	12/06/2026, 15:16:41
62.0889g CO ₂ e	[Log · Claude Code]	claude-sonnet-4-6	12/06/2026, 14:59:14
61.9543g CO ₂ e	[Log · Claude Code]	claude-sonnet-4-6	12/06/2026, 14:59:10
61.8333g CO ₂ e	[Log · Claude Code]	claude-sonnet-4-6	12/06/2026, 14:59:05
61.7549g CO ₂ e	[Log · Claude Code]	claude-sonnet-4-6	12/06/2026, 13:46:28
59.3041g CO ₂ e	[Log · Claude Code]	claude-sonnet-4-6	12/06/2026, 12:28:54

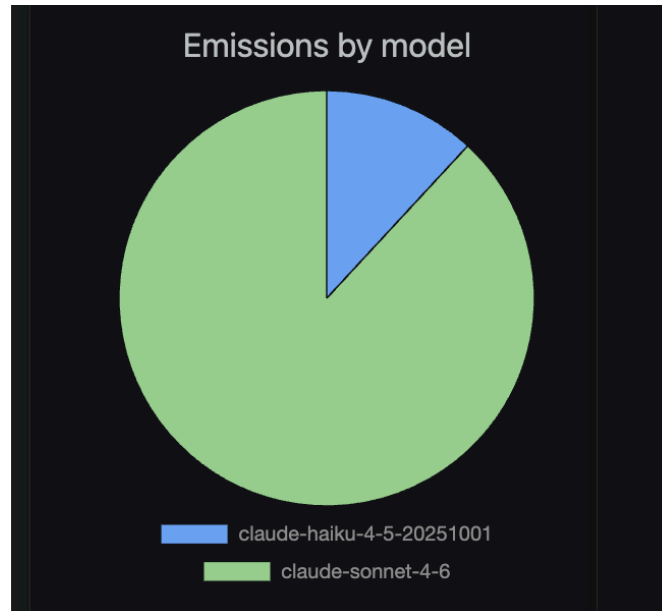
Mini View

The Mini View provides a lightweight summary of project activity directly within the VS Code Explorer sidebar. This allows users to monitor impact without opening the full dashboard.



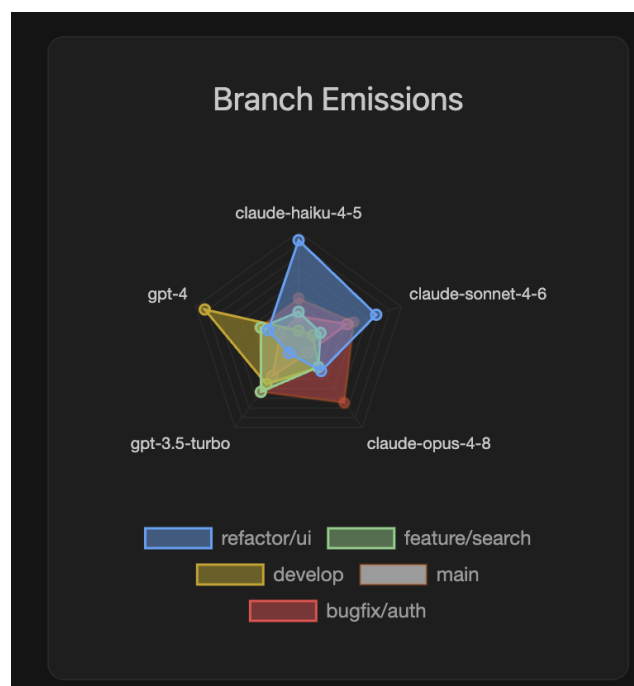
Model Breakdown

The model breakdown visualisation shows how estimated emissions are distributed across the models used within a project. This can help teams understand which models are responsible for the largest share of estimated emissions.



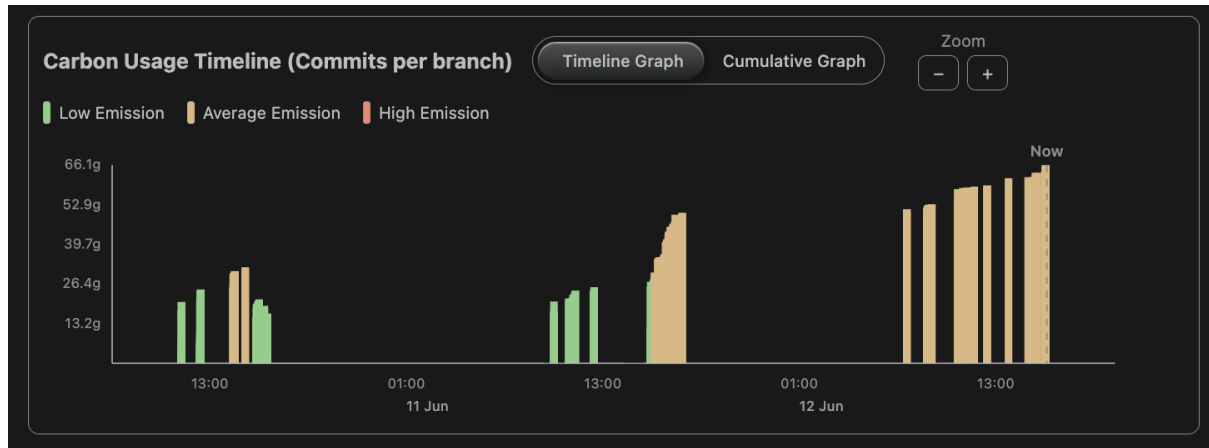
Branch Comparison

The branch comparison visualisation compares model usage across branches, helping teams identify differences in AI usage patterns between streams of work. This can be useful when exploring how different features, experiments, or approaches influence AI usage and estimated emissions.



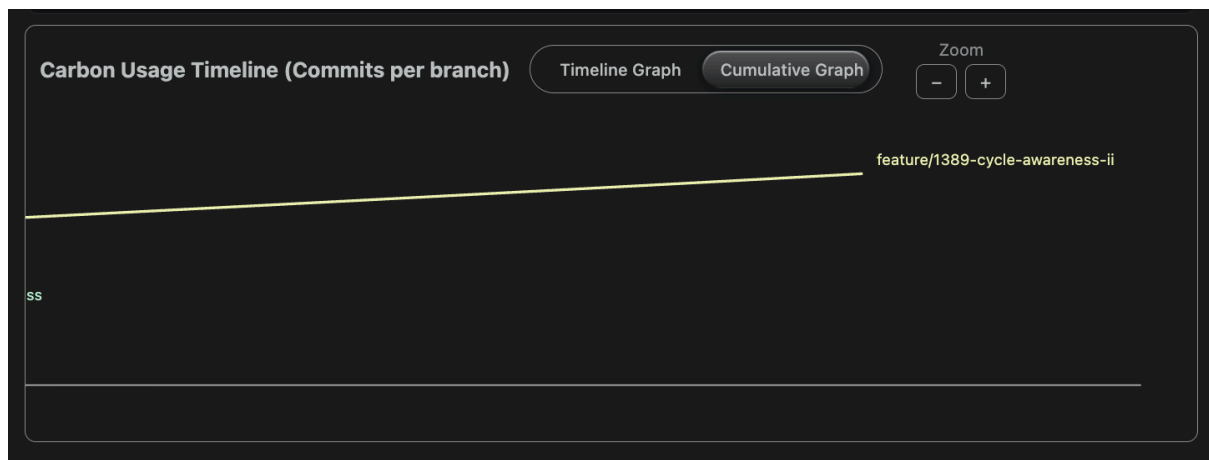
Timeline Analysis

Timeline views help users understand when AI activity occurs and identify periods of increased usage. This can be useful for understanding development patterns, release cycles, and experimentation periods.



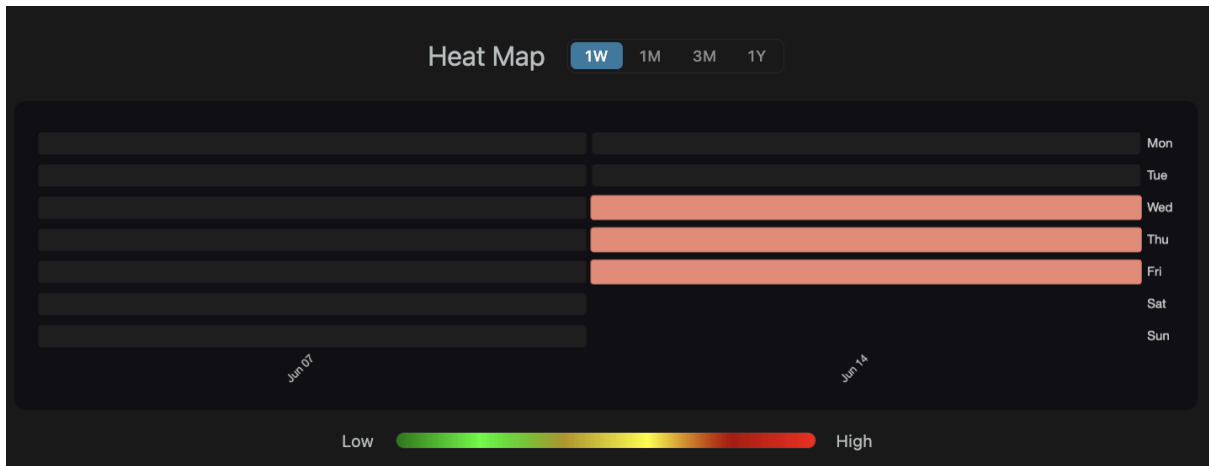
Cumulative Trends

The cumulative view shows how estimated emissions build up over time across a project or branch. This can help teams understand longer-term trends and compare the overall impact of different streams of work.



Historical Heatmap

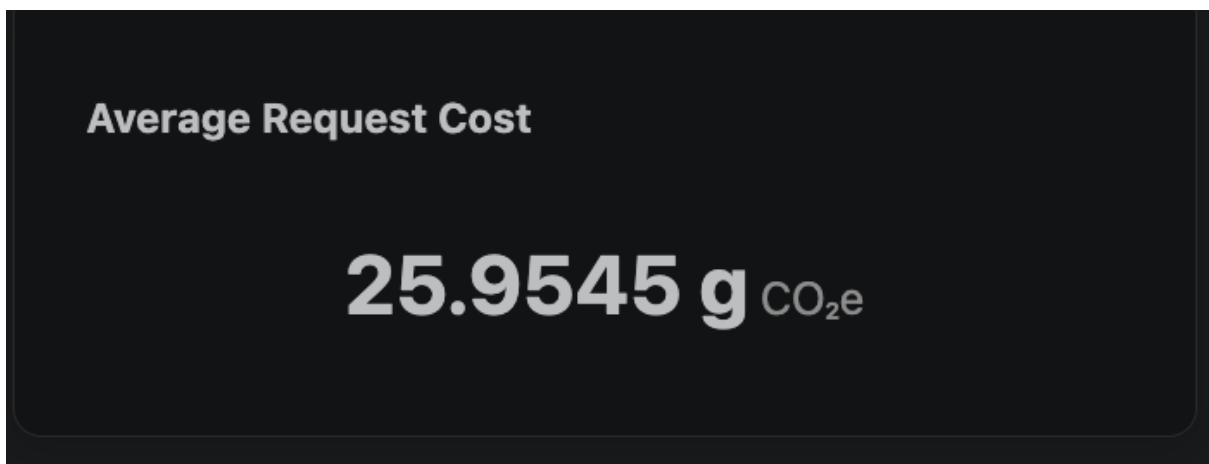
The heatmap provides a longer-term view of AI activity and estimated emissions across a project. This makes it easier to identify trends and recurring patterns over time.



Average Request Impact

The average request indicator shows the estimated impact of a typical AI interaction within the project.

This can help teams understand how individual requests compare with overall project activity and identify changes in usage patterns over time.

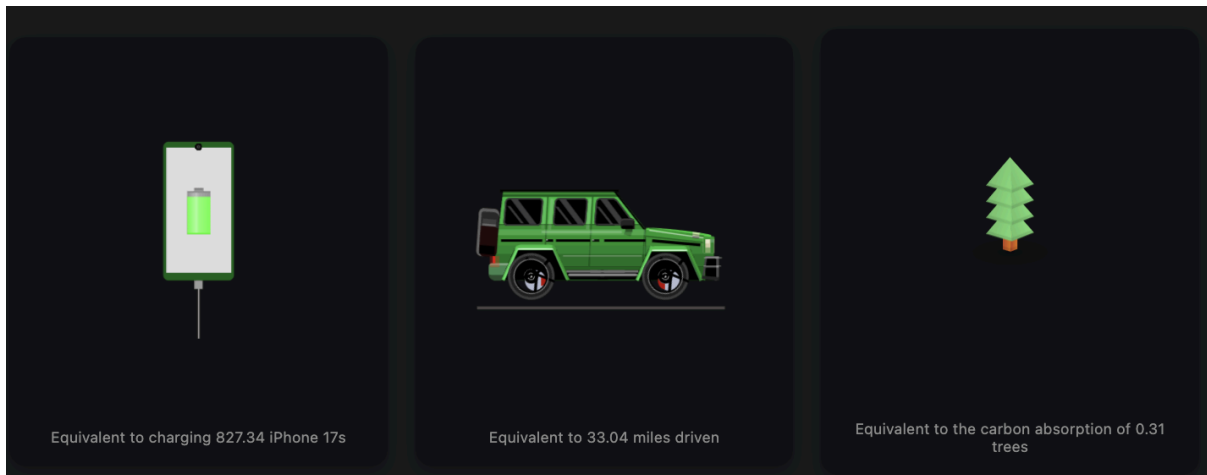


Real-World Equivalents

Carbon emissions can be difficult to interpret on their own.

To provide additional context, Prism translates estimated emissions into familiar real-world comparisons. These examples help users understand the relative scale of AI activity and build intuition about project impact.

The comparisons are intended as illustrative examples rather than precise equivalences.



Understanding Relative Impact

One of Prism's core features is Relative Impact Classification.

Every AI interaction is assigned a colour classification based on how it compares to other requests within the same project.

Classification	Percentile Range	Description
Green	Below the 50th percentile	Lower relative impact
Amber	Between the 50th and 90th percentile	Moderate relative impact
Red	Above the 90th percentile	Higher relative impact

These classifications compare requests against other requests recorded within the same project. To ensure meaningful comparisons, Relative Impact Classification is only applied after a minimum number of requests have been recorded. This threshold can be configured by the user.

For example, a request classified as red falls within the highest estimated-impact 10% of requests recorded in that project.

A red request does not mean a request is environmentally harmful.

A green request does not mean a request is environmentally sustainable.

The classifications simply help users identify requests that are unusually resource-intensive compared to the rest of the project. The goal is awareness rather than judgement.

7 Last Request: 66.1021 g CO₂e

Understanding Budgets

Prism includes optional project-level budgeting features.

Budgets are intended as awareness tools rather than targets.

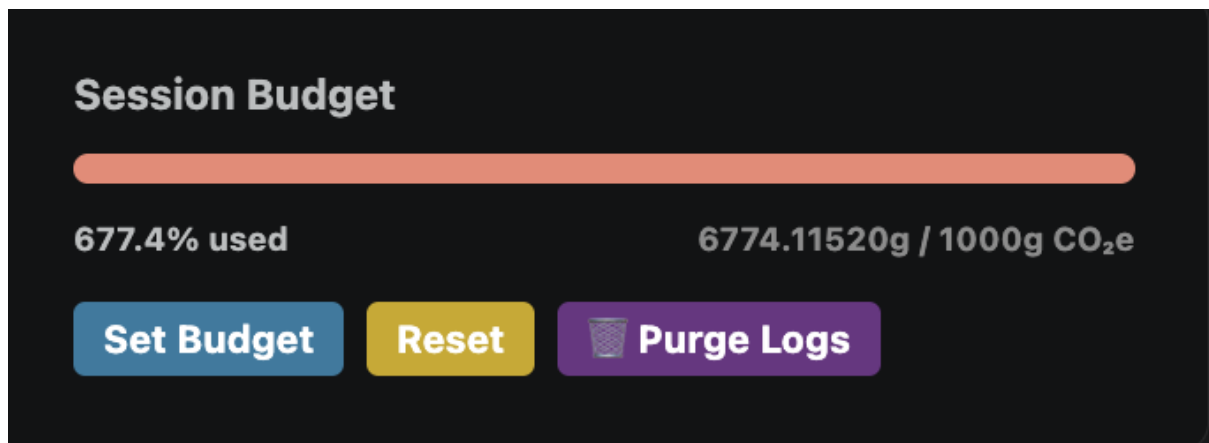
They can help teams:

- Establish a baseline
- Monitor cumulative impact
- Track changes over time

Budgets and Relative Impact Classification serve different purposes.

Relative Impact helps identify unusual requests.

Budgets help understand cumulative project impact.



Understanding the Journey

Prism is designed to support a gradual learning journey.

Observe

Start by using Prism without worrying about targets or budgets.

Focus on understanding:

- Which models are being used
- Which requests generate the highest impact
- When spikes occur during development

At this stage, the goal is simply to build visibility into AI usage patterns that are often hidden during software development.

Learn

As more interactions are recorded, Relative Impact Classification helps identify patterns and outliers. This can help teams understand where AI usage is concentrated and how different development practices affect estimated emissions.

Reflect

Once a project has established a baseline, teams can begin asking questions such as:

- Are we using the most appropriate model for the task?
- Are prompts becoming unnecessarily large?
- Are there opportunities to reduce repeated requests?
- Which activities generate the greatest impact?

The purpose is not to minimise AI usage at all costs, but to better understand where AI delivers value and where more efficient approaches may exist.

Improve

Over time, teams may choose to use Prism's budgeting and tracking features to establish their own benchmarks and monitor progress throughout a project.

Budgets are most useful once a team has developed an understanding of its typical usage patterns.

Rather than acting as sustainability targets, they provide a project-specific benchmark that can be used to track cumulative impact over time.

Privacy and Data Handling

Prism has been designed with privacy in mind.

Prism does not collect, transmit, store, or share:

- Source code
- Prompts
- AI conversations
- Personal information
- Usage data

Prism does not include:

- Telemetry
- Behavioural analytics
- User tracking
- Account monitoring

All analysis is performed locally using data already available on the developer's machine.

Any information displayed within Prism remains under the user's control.



Known Limitations

Prism provides directional estimates intended to support awareness and decision-making.

Results should not be interpreted as exact measurements of environmental impact.

Current limitations include:

- Image generation emissions are not yet supported
- Water consumption metrics are not included
- Estimates depend on publicly available model data
- Methodologies may evolve as new research becomes available

What Prism Does Not Do

Prism does not determine whether a project is sustainable.

Prism does not provide exact carbon measurements.

Prism does not tell users when they should or should not use AI.

Prism does not replace broader responsible AI practices.

Instead, Prism provides visibility into an area of software development that is often difficult to see, helping teams make more informed decisions and build a better understanding of the impact of AI.

Future Development

Future versions of Prism may include:

- Support for image generation workloads
- Water consumption metrics
- Support for additional AI development tools and model providers
- Improved model-specific estimation methodologies

Further Reading

For users who would like to explore the implementation in more detail:



README

Covers:

- Project overview
- Features
- Installation
- Running locally
- Architecture
- Repository structure
- Privacy
- Contributors
- Development setup
- Release process

Emission Calculation Methodology

Covers:

- Token-to-energy-to-carbon calculations
- Data sources
- Model conversion factors
- Assumptions and limitations
- Future methodology improvements

Contributors

Prism was originally developed through a collaboration between ustwo and students from the University of Bristol.

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University of Bristol Team

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Supporting Mentor

- Murray Groves

ustwo Contributors

Following delivery of the University of Bristol project, the following ustwo team members contributed to refining, stabilising, and improving the Prism prototype:

- Carlos Rodrigues
- Graham Richards
- Anjani Shingadia
- Ross Harrington